

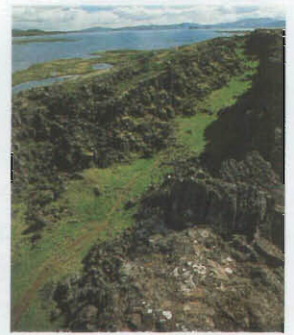
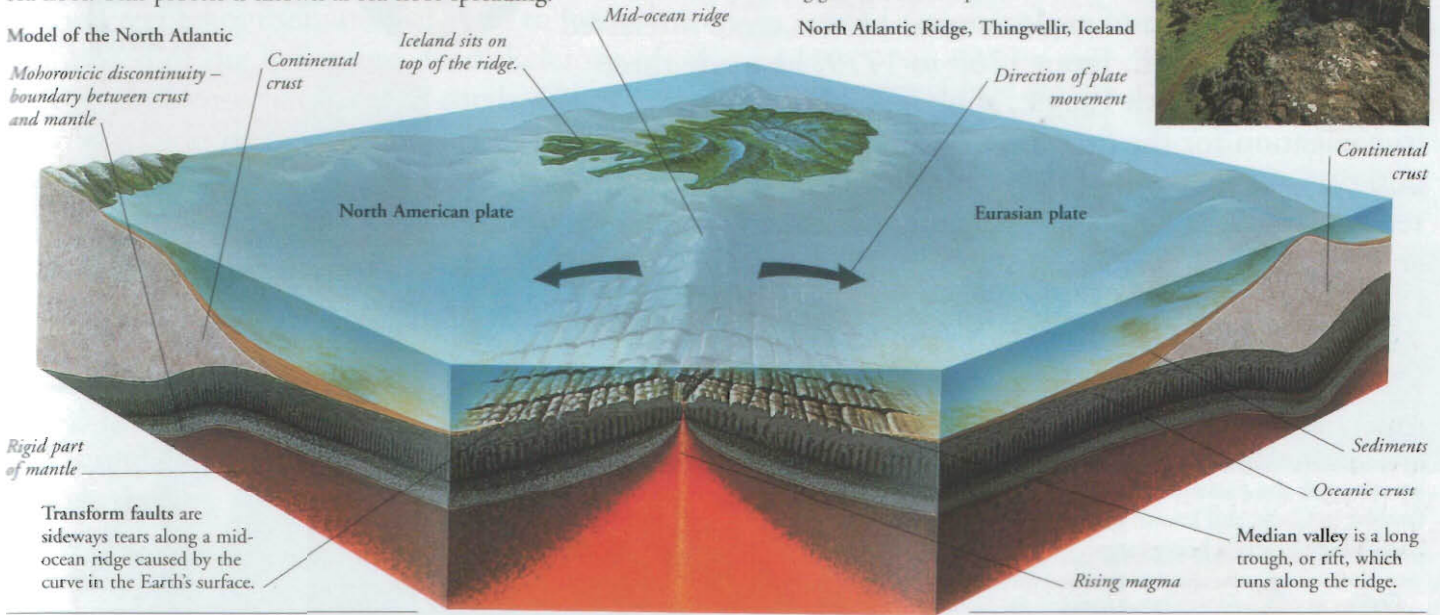
Diverging plates

At some places beneath the world's great oceans, the tectonic plates that make up the Earth's surface are slowly diverging. These places are called constructive margins. As the plates pull apart, molten rock called magma wells up through the crack between the plates and emerges as lava. When the lava cools, it adds new material to the sea floor. This process is known as sea-floor spreading.

Mid-ocean ridge

As two plates pull apart, the lava emerging from the Earth's interior solidifies and builds a line of undersea mountains along the crack. This is called a mid-ocean ridge. There is such a ridge beneath each of the world's great oceans. In Iceland, the North Atlantic Ridge rises above sea level and can be seen as a long gash in the landscape.

Model of the North Atlantic



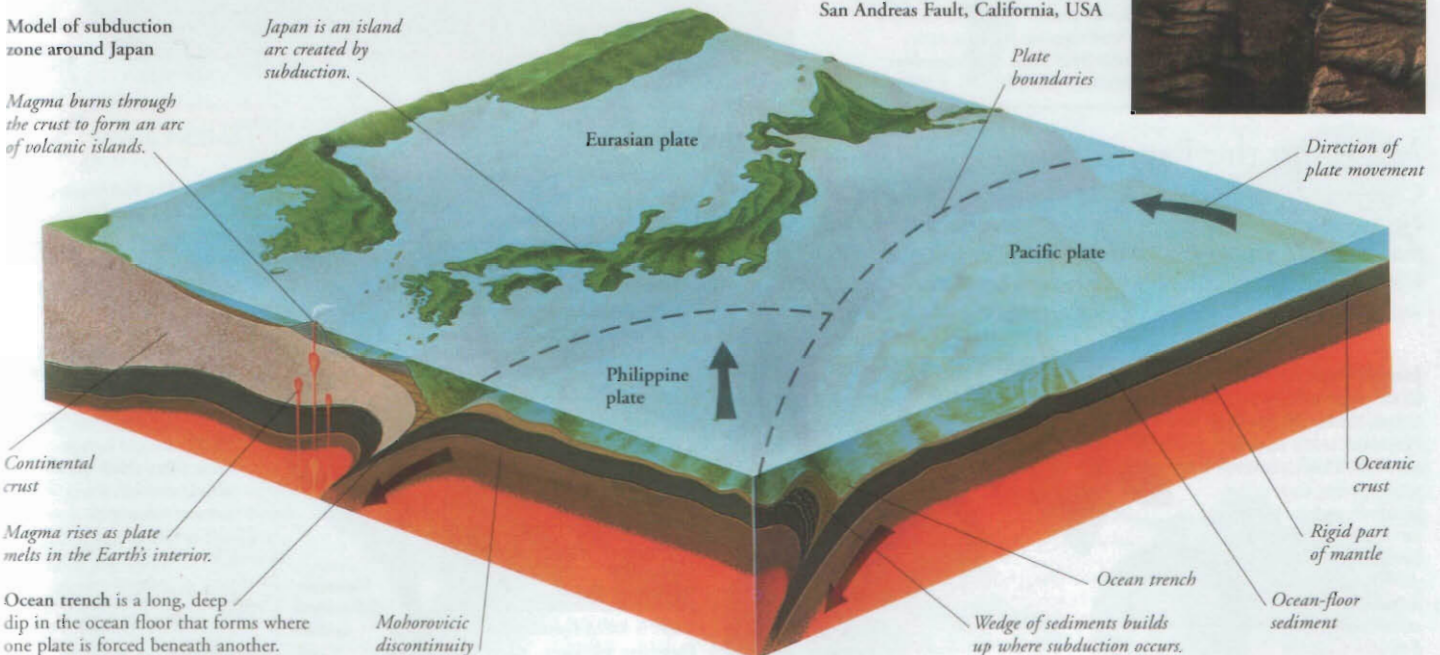
Converging plates

In many places the tectonic plates are converging, or moving against one another. In places known as destructive margins, oceanic plates are drawn underneath the continental plates. The oceanic plate is pulled down into the layer of the Earth's interior called the mantle, where it is destroyed by intense heat. This process is called subduction. In other places called collision zones, the edges of two continental plates may crumple as they collide. This process creates great mountain ranges such as the Alps and the Himalayas.

Transform

In some places where tectonic plates meet, the plates neither collide nor pull apart, but simply slide past each other in opposite directions. These places are called transforms, or conservative margins. Perhaps the most famous transform is the San Andreas Fault in California, USA, where the Pacific and North American plates grind slowly past one another. As the plates move, they often snag and judder, setting off violent earthquakes.

Model of subduction zone around Japan



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